LEF MODEL · CHAPTER 3

Chapter 3: The Engine Within — the Source Pressure of 150 GPa and the Foamed Magma

Based on Paper 03 — LEF Mechanism in Practice (Introduction and Part A: §A.1–A.13)

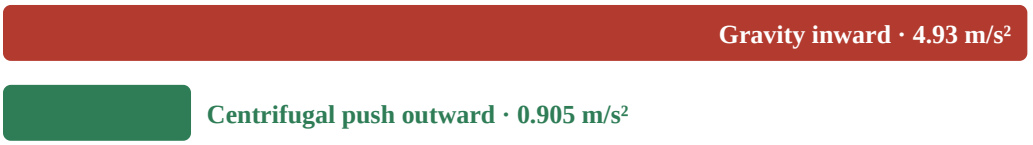
01 · THE MYSTERY

The mystery: who pushed 73 billion billion tonnes to a height of 16,000 kilometres?

In the previous chapters we saw that the corotation line solves the riddle of speed and the angular-momentum budget: if only we can bring the Moon's material to a distance of about 16,000 km from the Earth's centre while it turns with it, it will already be moving at the speed of a free orbit. But now we face the hardest question of reality:

> What force in the world can lift an enormous mass of 7.35×10^{22} kg against the Earth's gravity?

To understand how demanding this task is, let us first examine two tempting but failing proposals. First, **rotation alone lifts nothing**: even on the surface of an Earth that turns every 5.5 hours, at a height of 8,991 km from the centre (a body of the Moon's mass touching the surface), gravity pulls inward with an acceleration of 4.93 m/s², while the centrifugal push outward is only 0.905 m/s². Every kilogram still weighs downward about forty per cent of its weight on the Earth's surface today! Rotation does lighten the weight, but it does not carry it.



At 8,991 km from the centre, with a 5.5-hour day: gravity is more than five times the centrifugal push.

Second, **buoyancy does not work in space**: oil floats on water because the heavier water around it pushes it up, but above the Earth's surface there is no ocean of rock to float the Moon upward.

The conclusion

The conclusion is clear: to build a moon from the Earth's own material, **an enormous internal pressure engine** is needed, working from within — an engine that pushes a column of magma upward from the Earth's depths, supplies mass at a sufficient rate, and does not collapse under the weight of the moon being built at its top.

02 · THREE IMAGES

An image: the toothpaste, the water that rises to the roof, and the soda bottle being opened

To get a hands-on feel for how this engine works, put together three simple images from everyday life:

01 · Source pressure

The toothpaste tube

Deep inside the young Earth, the weight of all the layers above creates tremendous pressure. When a flow path opens from the depths to the surface in one favoured region, the internal pressure squeezes the molten rock upward — just as toothpaste comes out of the nozzle when you press the tube.

02 · Hydrostatic head

A pump that lifts water to a tank on the roof

For a pump in the basement to send water up to a tank on the roof of a skyscraper, its pressure must be greater than the weight of the whole column of water inside the pipe. If the water is too heavy and the building too tall, the column stops halfway.

03 · Foamed magma

A soda bottle being opened

At great depth, under enormous pressure, water and volatile gases are dissolved in the molten rock like gas in a closed soda bottle. As soon as the magma rises and the pressure on it falls, the gases come out and form billions of bubbles — **foamed magma**. This foam is much lighter than solid rock, so exactly the same pressure can lift it much higher!

03 · SCIENCE AND MODEL

Science and model: why dry rock fails — and only volatile-rich magma passes the threshold

In the LEF model we do not guess that the pressure is enough — we **price the push** from the very same gravitational field. Before calculating, Paper 03 sets a strict bookkeeping rule: at every moment, the initial mass of the parent body ($M_0 = 6.0455 \times 10^{24}$ kg) is divided into four clear reservoirs, so that no kilogram is counted twice:

M_E	m	M_{transit}	M_{escaped}
The mass remaining in the Earth	The mass accumulated in the developing body	The mass in motion on the way between them or in ejection	And the mass lost from the system

Now the paper sets up a one-dimensional pressure test along the magma column, from the centre of the young Earth ($R_V = 6{,}517$ km, a 5.5-hour day) to the corotation radius of the whole parent body ($r_* = 15{,}882.9$ km):

The pressure available at the source p_s: the weight of the layers pressing on the centre of the young Earth (minus the centrifugal relief of the fast rotation) gives a lithostatic source pressure of **150.26 GPa** (about one and a half million atmospheres!).

The cost of the magma column p_{head}: to hold up a magma column made of deep magma at a density of $\rho_d = 3{,}500$ kg/m³ that foams above the surface to a density of $\rho_o = 1{,}500$ kg/m³, a pressure of **139.78 GPa** is needed.

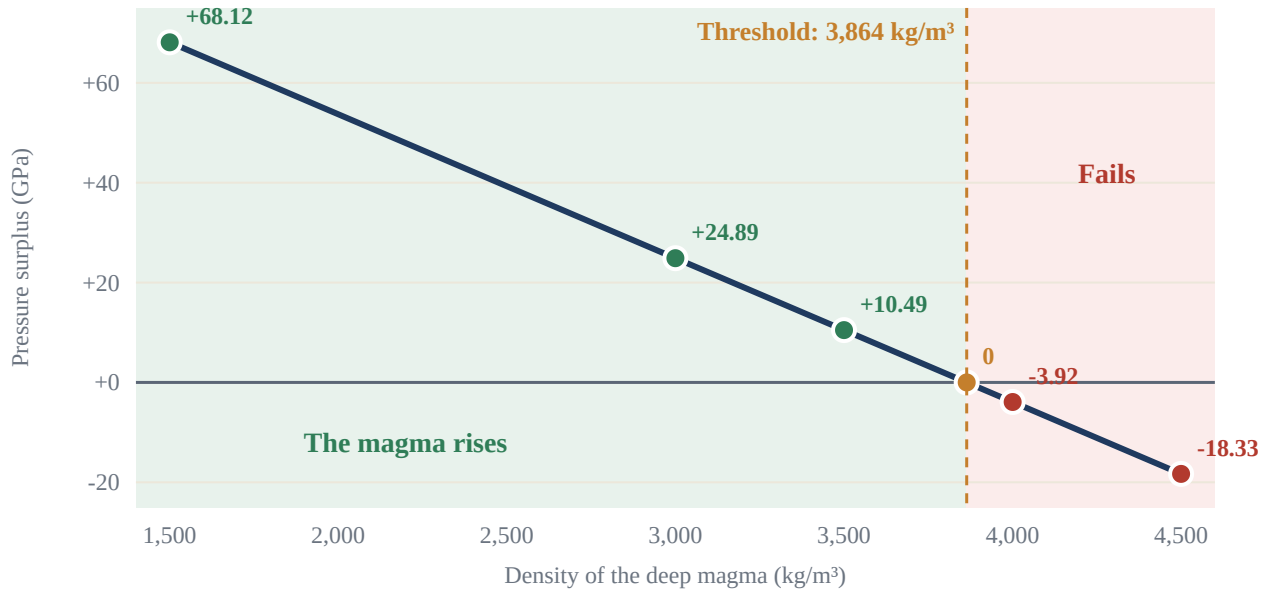
The surplus that pushes the Moon: 150.26 − 139.78 = **+10.49 GPa**! A positive pressure surplus of more than ten gigapascals remains, driving the flow upward.



The pressure balance at the reference density of 3,500 kg/m³: the source pressure covers the weight of the magma column and leaves a surplus.

`The critical density threshold · 3,864 kg/m³`

When we check which magma densities pass the test, a sharp boundary appears: any deep magma with a density below **3,864 kg/m³** (like magma rich in water and volatiles) leaves a positive pressure surplus and reaches the corotation line; by contrast, a dry, heavy magma at a density of 4,000 kg/m³ requires 154.19 GPa and fails with a deficit of −3.92 GPa. **The water and the volatiles are not an incidental addition — they are the physical condition that allows the Moon to rise!**



The pressure surplus (before load and friction) as a function of the density of the deep magma, according to Paper 03: the line crosses zero at 3,864 kg/m³.

04 · THE MECHANISM IN WORDS

The mechanism in words: the flow rate, the energy source and the rejection of the thin pipe

We have found that a pressure surplus of 10.49 GPa can push a column of foamed magma all the way to the corotation line. But to build a whole moon, the internal engine must also pass **three more practical tests** within the very same system:

TEST 1

The rate test

Does the magma arrive in time? To deliver the mass of a whole moon (7.35×10^{22} kg) over the building window of 600 million years (or 100 million years in the sensitivity test), an average rate of 3.88×10^{6} kg/s is needed (and up to 2.33×10^{7} kg/s in the short window). By the law of viscous flow (Poiseuille), the flow rate grows as **the fourth power of the passage radius** (b^4): a passage with a radius of several hundred kilometres or more carries the mass easily, as long as its interior stays hot and mobile.

TEST 2

The energy test

Where does the work of lifting a moon come from? Lifting the Moon's mass from the surface (8,991 km) to the release radius (16,400 km) requires gravitational work of about 1.47×10^{30} **joules** (less than one per cent of the Earth's binding energy). Where does this energy come from? From the very same process that drives the internal evolution: **the sinking of heavy iron into the Earth's core** and the contraction of the young Earth release gravitational energy of about 10^{31} joules — more than enough to fund the lift!

TEST 3

The insulating-crust test

How does the passage avoid freezing on the way? While the interior of the passage is fed with heat from below and stays liquid, its outer surface freezes on contact with the cold surroundings. The equation of the freezing front shows that the first metre of the outer crust solidifies within only **1.9 to 23 days**, one kilometre within thousands to tens of thousands of years, while a thick crust of 50 km takes millions of years — so a solid, insulating shell forms that keeps the inner core hot and flowing.

The surprising discovery of the combined test

Why a "thin pipe" is ruled out entirely

When the pressure, the flow and the Moon's weight are checked together in the last climbing stage (a 5.77-hour day, latitude 22.5°, the Moon's centre at 15,397 km), a dramatic result appears: if the passage were a narrow pipe with a radius of 500 km, the weight of the Moon pressing on the tip of the pipe would create a load of 46.18 GPa and wipe out the whole pressure surplus into a deficit of −35.03 GPa. The radial balance turns positive only above **a passage radius of 1,017 km** — and at such a radius, the outer part of the passage itself already weighs **0.46 of the Moon's mass** (and at a radius of 1,500 km — a whole Moon's mass!) and carries a sideways shear stress of gigapascals. **The conclusion:** the passage is not a massless thin pipe, but a wide, hot and massive base of the Earth itself — **the pear**!

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The pressure balance along the feeding route connects the lithostatic source pressure at the centre (p_s), the hydrostatic head of the deep (I_d) and outer (I_o) magma column, the weight load of the body at the top of the column (p_{load}), and the viscous friction losses (Δp_{loss}):

The source pressure

$$p_s = \frac{\rho_p}{2}\left(\frac{GM_0}{R} - \omega_0^2 R^2\right) = 150.26\ \text{GPa}$$

The cost of the magma column

$$p_{\text{head}} = \rho_d I_d + \rho_o I_o$$

The lifting condition and the density threshold

$$p_s - p_{\text{head}} - p_{\text{load}} - \Delta p_{\text{loss}} > 0 \implies \rho_{d,\max}$$

For laminar flow in a passage of radius b and length ℓ (Poiseuille's law), and for the thickening rate of the freezing front of the outer crust (h_{fr}) and the gravitational work of the lift (W_g):

The flow rate (Poiseuille)

$$Q = \frac{\pi b^4 \Delta p}{8\eta\ell}$$

The freezing front

$$h_{\text{fr}}(t) \simeq \sqrt{\frac{2k\,\Delta T\,t}{\rho_s L_f}}$$

The work of the lift

$$W_g = GM_E m\left(\frac{1}{r_0} - \frac{1}{r_1}\right) \simeq 1.47 \times 10^{30}\ \text{J}$$

06 · THE SIGNATURE

The signature in reality: the melt inclusions inside the

lunar crystals

The physical test established that only magma rich in water and volatiles (lighter than 3,864 kg/m³) can pass the pressure threshold of 150.26 GPa and rise to the corotation line. Had the Moon formed from a dry, scorching giant impact that vaporised the water, its ancient rocks should have been completely dry.

But when scientists examined, with advanced microscopy, **melt inclusions** — tiny droplets of ancient magma trapped inside olivine crystals that grew deep in the Moon and protected the water from escaping to space — a surprising signature was found: a high water content of **615 to 1,410 ppm**, similar to the concentrations in the Earth's upper mantle.

> The water found inside the lunar crystals is the fossil remnant of that ancient "foaming engine" that pushed the Moon upward!

The lab box: the pressure balance, the density threshold and the combined test (Paper 03, Part A)

Density of the deep magma ρ_d	Available source pressure p_s	Cost of the magma column p_{head}	Pressure surplus before load and friction	Result of the lifting test
1,500 kg/m³	150.26 GPa	82.14 GPa	+68.12 GPa	Passes easily (strongly foamed magma)
3,000 kg/m³	150.26 GPa	125.37 GPa	+24.89 GPa	Passes (volatile-rich magma)
3,500 kg/m³ (reference)	**150.26 GPa**	**139.78 GPa**	**+10.49 GPa**	**Passes — the surplus drives the building**
3,864 kg/m³ (threshold)	**150.26 GPa**	**150.26 GPa**	**0.00 GPa**	**The exact point of balance**
4,000 kg/m³	150.26 GPa	154.19 GPa	−3.92 GPa	Fails (magma too dry and heavy)

Passage radius b	Radial load of the Moon	Flow friction loss	Radial surplus	Sideways shear stress (22.5°)	Mass of the outer passage (in Moon masses)
500 km	46.18 GPa	0.0103 GPa	−35.03 GPa (fails)	46.04 GPa	0.112
1,000 km	11.54 GPa	0.0006 GPa	−0.39 GPa (threshold: 1,017 km)	11.51 GPa	0.448 (almost half a Moon!)
1,500 km (the contact disc)	**5.13 GPa**	**0.0001 GPa**	**+6.02 GPa (passes!)**	**5.12 GPa**	**1.008 (a massive structure — the pear)**

07 · NEXT

The door ahead

We have discovered how the engine within works: the deep source pressure of 150.26 GPa lifts foamed magma lighter than 3,864 kg/m³, and the combined test revealed that there is no thin pipe here — but a wide, massive passage rising out of the Earth.

But what happened to the Earth's outer crust that lay right above that building zone? In the next chapter — **Chapter 4: The Skin That Rises from Outside — the 4 Stages of the Dome and the Translation Rule of the Moon's Face** (Paper 03, Part B: §B.1–B.7) — we will discover how the Earth's sial shell rose with the body, became the Moon's skin, and divided the Moon's face into three precise geographic zones.

Chapter 4

The Skin That Rises from Outside — the 4 Stages of the Dome and the Translation Rule of the Moon's Face →

← Chapter 2 · Prologue

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